A MOD 32 COUNTEREXAMPLE TO THE CONGRUENCE FORM OF PROPP'S CONJECTURE 2 FOR AZTEC DIAMOND TILINGS BY DOMINOES AND SQUARE TETROMINOES

Abstract. Let $M(n)$ be the number of tilings of the Aztec diamond of order n by dominoes and square tetrominoes (OEIS A356512). Conjecture 2 of Propp [*Integers* **23** (2023), Art. A30] asserts that for all $k \geq 1$, $n + n' \equiv -3 \pmod{2^k}$ implies $M(n) + M(n') \equiv 0 \pmod{2^k}$. We refute this congruence form at the single instance $(n, n', k) = (14, 15, 5)$: here $14 + 15 = 29 \equiv -3 \pmod{32}$, while

$$M(14) \equiv 79, \qquad M(15) \equiv 225 \pmod{256},$$

so $M(14)+M(15) \equiv 48 \pmod{256}$, whose 2-adic valuation is 4, not ≥ 5. Propp's Conjecture 1, on the 2-adic continuity of M, is not refuted, and neither is the antisymmetry restatement of Conjecture 2, which presupposes it.

1. The conjecture

For $n \geq 0$ let AD_n be the Aztec diamond region of order n: the union of the closed unit squares of the integer lattice that lie entirely inside $|x|+|y| \leq n+1$. Equivalently AD_n has $2n$ rows, the row r $(0 \leq r \leq 2n-1)$ consisting of $2\min(r+1,\, 2n-r)$ cells centred on the vertical axis, so that

$$|AD_n| \;=\; 2n(n+1). \tag{1}$$

Let

$$M(n) \;=\; \#\{\text{tilings of } AD_n \text{ by } 1\times 2,\ 2\times 1 \text{ and } 2\times 2 \text{ tiles}\},$$

that is, by dominoes in either orientation together with the square tetromino. Then $M(0) = 1$, $M(1) = 3$, and M is the sequence OEIS **A356512**. Propp [3] states two conjectures about the 2-adic behaviour of M; the second, quoted here from p. 6 of the published version, is the object of this note.

Conjecture 2 (Propp [3]). *For all $k \geq 1$, if $n + n' \equiv -3 \pmod{2^k}$ then $M(n) + M(n') \equiv 0 \pmod{2^k}$.*

Both n, n' range over $\mathbb{Z}_{\geq 0}$ and k over $\mathbb{Z}_{\geq 1}$; there is no existential quantifier, so a single triple (n, n', k) satisfying the hypothesis and violating the conclusion refutes the statement. Propp prints $M(n)$ for $0 \leq n \leq 12$ and remarks that he has no efficient method of computing further terms; since $n, n' \leq 12$ forces $n + n' \leq 24 < 29$, that table cannot test $k = 5$ at any point.

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2. The counterexample

Theorem 1. *The congruence form of Conjecture* 2 *of* [3], *as quoted in §1, is false. Explicitly, take* $n = 14$, $n' = 15$ *and* $k = 5$. *Then* $n + n' \equiv -3 \pmod{2^k}$, *while* $\mathrm{v}_2(M(14)+M(15)) = 4$ *and so* $M(n)+M(n') \not\equiv 0 \pmod{2^k}$.

Proof. The two residues, computed as described in §3, are

$$M(14) \equiv 79 \pmod{256}, \qquad M(15) \equiv 225 \pmod{256}. \tag{2}$$

The hypothesis holds: $14 + 15 = 29$ and $29 + 3 = 32 = 2^5$, so $14 + 15 \equiv -3 \pmod{32}$; equivalently $(-3 - 14) \bmod 32 = 15 = n'$. Both indices are nonnegative and $k = 5 \geq 1$, so the triple lies in the range quantified over.

The conclusion fails. By (2), $M(14) + M(15) \equiv 79 + 225 = 304 \equiv 48 \pmod{256}$, and $48 = 2^4 \cdot 3$. A residue modulo $256 = 2^8$ determines the 2-adic valuation of an integer whenever that valuation is less than 8, so $\mathrm{v}_2(M(14) + M(15)) = \mathrm{v}_2(48) = 4$. In particular $M(14) + M(15) \equiv 16 \not\equiv 0 \pmod{32}$. The valuation is short of the required 5 by exactly one power of two. □

3. The computation and its controls

The residues (2) were obtained by a dense broken-profile transfer computation over the $2n \times 2n$ bounding box of AD_n, one cell at a time, with $2n$ frontier bits and one extra bit for the cell that a 2×2 tile placed at the previous column occupies on the next row; cells outside AD_n are treated as permanently blocked, and the programs assert that the number of admissible cells equals $2n(n + 1)$ as in (1). At $n = 14$ and $n = 15$ this carries 2^{29} and 2^{31} states. Two implementations written independently of each other, in C, agreed on both residues, and each residue was produced twice, on different machines. A separate corroboration of $M(14)$ is arithmetic: an exact accumulator modulo 2^{128} returned $M(14) \equiv 228062845944308165005028192227771743567 \pmod{2^{128}}$, and since 2^8 divides 10^8 the last eight decimal digits of that number determine the residue modulo 256, giving $71743567 = 280248 \cdot 256 + 79$, in agreement with (2).

A third implementation accompanies this note as `verify.py`: a pure Python program, standard library only, exact integer arithmetic, sharing no code with either C engine. It recomputes AD_n from the inequality $|x| + |y| \leq n + 1$ and checks (1); reproduces $M(0), \ldots, M(4)$ by plain backtracking with no transfer computation at all; reproduces all thirteen published terms of A356512 exactly; reproduces the domino-only specialisation $2^{n(n+1)/2}$ of Elkies, Kuperberg, Larsen and Propp [1] (OEIS A006125) for $n \leq 12$ as a control on the region and the domino rules; then recomputes $M(14)$ and $M(15) \bmod 256$ from scratch, obtaining 79 and 225, and rederives Theorem 1 from its own residues rather than from the printed ones. Its recorded run, `verify.output.txt`, reports `VERDICT: ALL 63 CHECKS PASS`. The deep part is expensive for an interpreted language: in that run the two residues took 777 and 1835 seconds on one core and the process peaked at 9.81 GB.

Two remarks of that transcript overreach — its check `E8` and its closing scope note — and nothing here relies on either.

4. Scope

Only the congruence form of Conjecture 2, exactly as printed and quoted in §1, is refuted. Propp restates it as: *"if one extends $M : \mathbb{N} \to \mathbb{N}$ to the 2-adic function $\widehat{M} : \mathbb{Z}_2 \to \mathbb{Z}_2$, one has $\widehat{M}(-3-n) = -\widehat{M}(n)$"*; that form presupposes the 2-adic extension, i.e. his Conjecture 1, which is neither used nor refuted here. $M(14)$ and $M(15)$ are established modulo 256 only; the exact integers are not computed anywhere here, and Theorem 1 does not need them. Nothing is claimed for an index ≥ 16: every other pair on the diagonal $n+n' = 29$ has an index ≥ 16, where M is unknown to us, so those instances are settled in neither direction, no instance with $k \geq 6$ was tested, and $(14, 15, 5)$ is not claimed to be the least failure. Finally, every control listed in §3 lies at $n \leq 12$, the range where published data exists; at $n = 14$ and $n = 15$ there is nothing external to compare against, and what stands in place of a control there is the agreement of the three independently written programs.

References

[1] N. Elkies, G. Kuperberg, M. Larsen, and J. Propp, *Alternating-sign matrices and domino tilings I*, J. Algebraic Combin. **1** (1992), no. 2, 111–132. doi:10.1023/A:1022420103267. The specialisation used as a control here, $2^{n(n+1)/2}$ domino tilings of AD_n, is OEIS A006125.

[2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequence A356512 (M, contributed by J. Propp, August 2022; thirteen terms, $n = 0, \dots, 12$) and A006125.

[3] J. Propp, *Some 2-adic conjectures concerning polyomino tilings of Aztec diamonds*, Integers **23** (2023), Art. A30; arXiv:2204.00158. Conjecture 2 is quoted from the published version, p. 6.